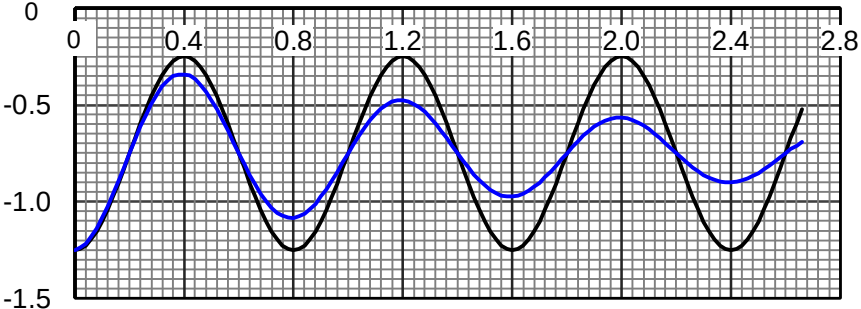


	$\approx 7.50 \times 10^{-3} \text{ J}$
4(a)(i)	Amplitude $x_0 = 0.50 \text{ cm}$ <i>Note: When reading Fig. 4.1, the value should be read to the appropriate number of decimal places of two, instead of one.</i>
4(a)(ii)	3 cycles of the oscillations take 2.4 s. The period is 0.80 s. Angular frequency $\omega = \frac{2\pi}{T}$ $= \frac{2\pi}{0.80}$ $= 7.854$ $\approx 7.85 \text{ rad s}^{-1}$
4(a)(iii)	Maximum speed $v_0 = x_0 \omega$ $= 0.500 \times 7.854$ $= 3.927$ $= 3.93 \text{ cm s}^{-1}$

No.	Solution
4(b)	The addition of a light piece of card causes light damping to the oscillations of mass M. Light damping causes the amplitude of the oscillation to continuously decrease exponentially with time, but does not increase the period of the oscillation.  Fig. 4.2
5(a)	Diffraction is the bending or spreading of waves when they travel through a